FDS Lab CA Task

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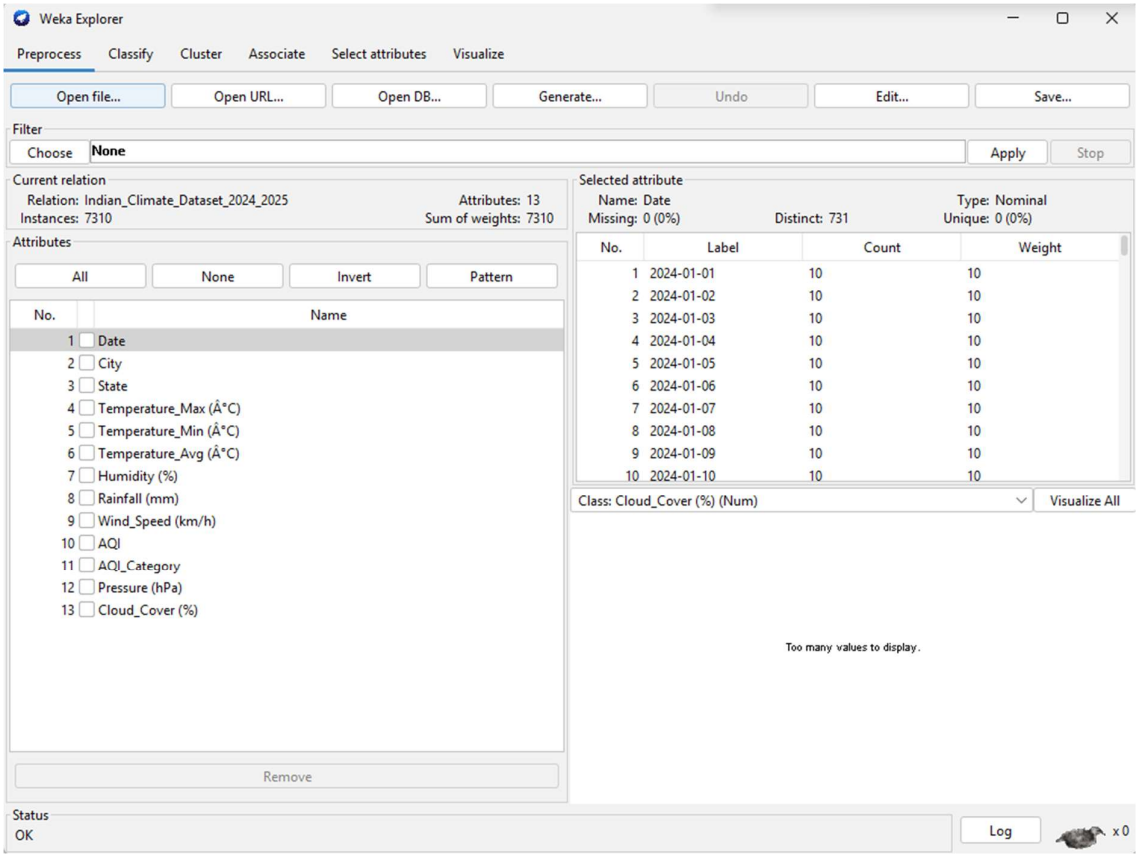
AI&DS A1

Dataset: https://www.kaggle.com/datasets/ankushnarwade/indian-climate-dataset-20242025

AI use case - KJS-CES-01: Climate Intelligence for Heatwave Monitoring, Prediction, and Early Warning

Using the Weka Tool, I performed the following tasks on my dataset:

1) Loading and inspecting



Instances: 7310

Attributes: 13

Missing Values: 0

Target attribute: AQI_Category

Target classes: Good, Satisfactory, Moderate, Poor, Very Poor

2) Checking for missing values

After applying “ReplaceMissingValues” Filter, instance count remains at 7310. Hence no missing values proved.

3) Removing irrelevant attributes

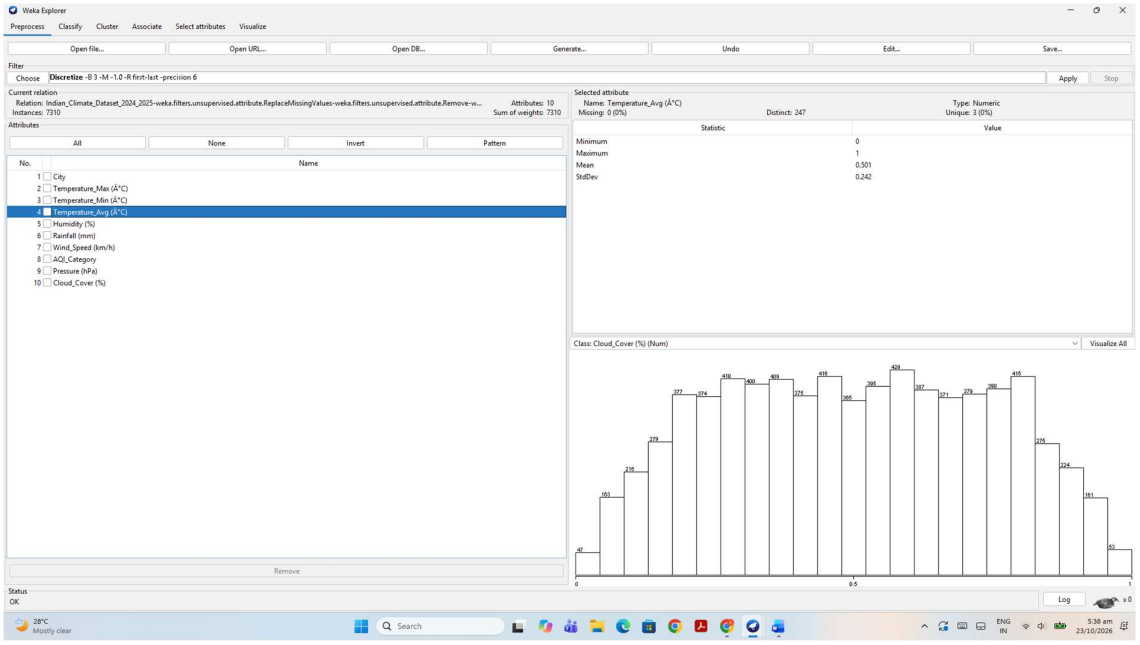
We remove the following attributes:

- Date: it should not be used as a predictor
- State: redundant with City
- AQI: avoid influencing predictions of AQI category

4) Normalising attributes

Normalised all numerical attributes to 0 to 1 scale from cleaned data

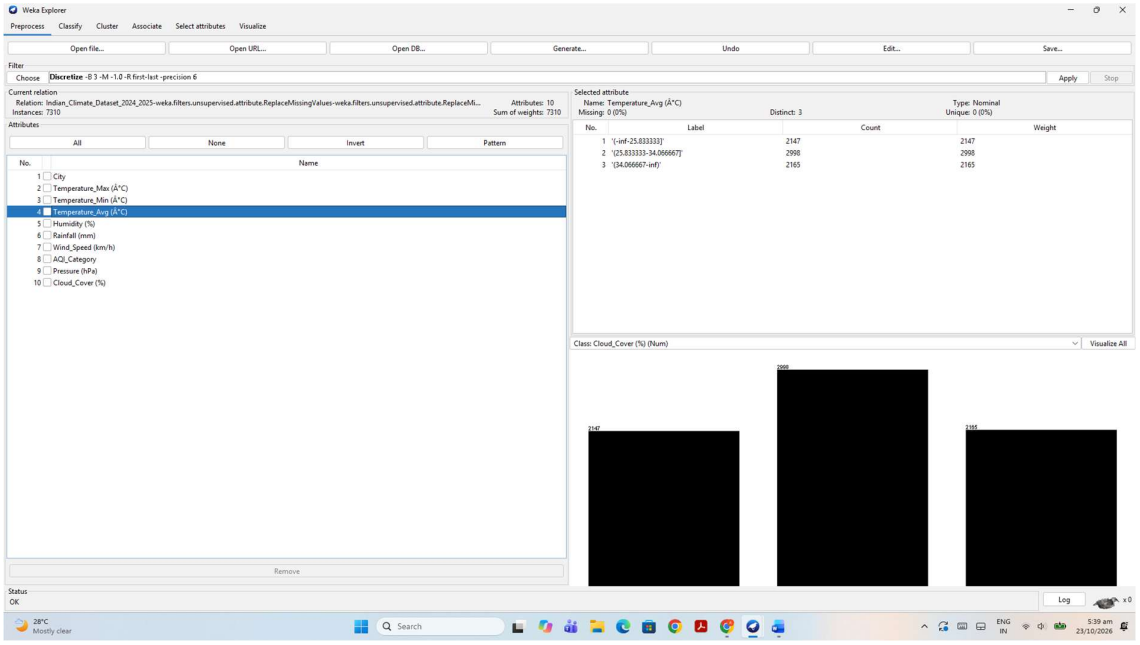
Eg:



5) Discretizing into bins

Creating 3 discrete bins from cleaned data

Eg:



6) Classification

Using J48, a decision tree tool in Weka, we base it off of the “AQI_Category” attribute.

```
Number of Leaves  :     723

Size of the tree :      1437


Time taken to build model: 0.11 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        2018               27.606  %
Incorrectly Classified Instances      5292               72.394  %
Kappa statistic                         -0.02
Mean absolute error                      0.3013
Root mean squared error                  0.4614
Relative absolute error                101.9471 %
Root relative squared error            120.0479 %
Total Number of Instances             7310

=== Detailed Accuracy By Class ===

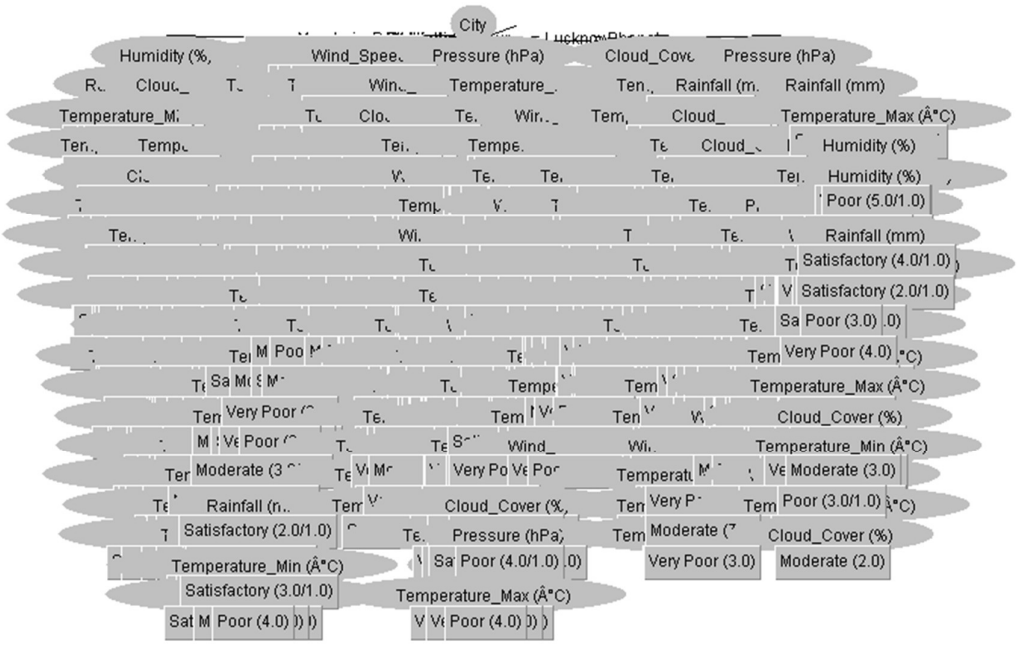
                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.394    0.422    0.306      0.394    0.345      -0.026   0.476     0.303     Poor
                 0.379    0.403    0.313      0.379    0.343      -0.022   0.480     0.312     Moderate
                 0.084    0.099    0.142      0.084    0.105      -0.019   0.479     0.155     Satisfactory
                 0.074    0.083    0.141      0.074    0.097      -0.011   0.497     0.153     Very Poor
                 0.020    0.014    0.049      0.020    0.028      0.010    0.517     0.040     Good
Weighted Avg.    0.276    0.296    0.247      0.276    0.256      -0.020   0.483     0.250

=== Confusion Matrix ===

    a    b    c    d    e   <-- classified as
  924  946  239  200   35 |    a = Poor
 1023  905  232  192   33 |    b = Moderate
  495  485  100   99   15 |    c = Satisfactory
  484  450  104   84   14 |    d = Very Poor
   93  103   31   19    5 |    e = Good
```

We get a very large decision tree, with a very poor rate of correctly classified instances. The confusion matrix showed that the model frequently confused the Poor and Moderate categories and performed particularly poorly on the Good category. The Kappa statistic of −0.02 indicated very weak agreement beyond chance. These results suggest that the classifier requires further investigation and improvement before it can be considered reliable for predicting AQI categories.

Decision tree:


City
Humidity (%,
Wind_Speed
Pressure (hPa)
Cloud_Cover
Pressure (hPa)
Rainfall (mm)
Temperature_Max (Â°C)
Humidity (%)
Humidity (%)
Poor (5.0/1.0)
Rainfall (mm)
Satisfactory (4.0/1.0)
Satisfactory (2.0/1.0)
Poor (3.0) .0)
Very Poor (4.0) .°C)
Temperature_Max (Â°C)
Cloud_Cover (%)
Temperature_Min (Â°C)
Moderate (3.0)
Poor (3.0/1.0) .°C)
Cloud_Cover (%)
Satisfactory (2.0/1.0)
Moderate (
Temperature_Min (Â°C)
Poor (4.0/1.0) .0)
Very Poor (3.0)
Moderate (2.0)
Satisfactory (3.0/1.0)
Temperature_Max (Â°C)
Poor (4.0)))
Poor (4.0)))